Crime in Times of Strike

Evidence from the 2012 Chicago Public Teacher Strike

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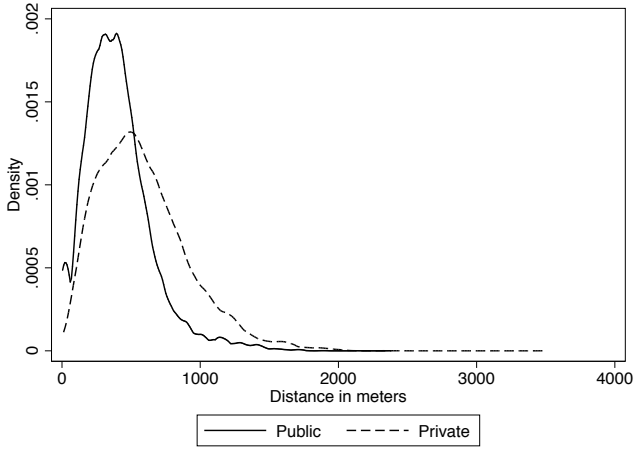
This paper

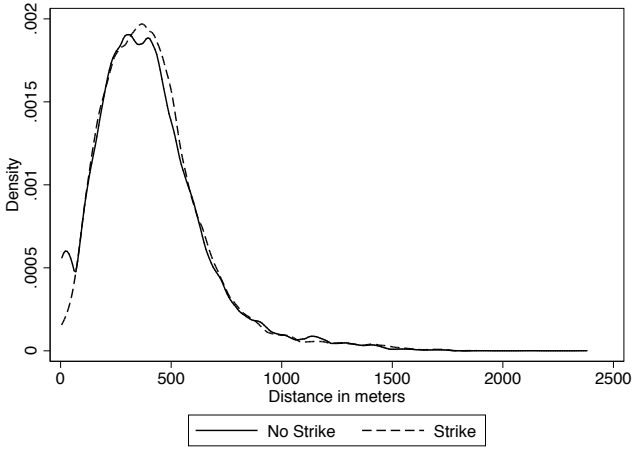
- Use 2012 Chicago public teacher strike as Natural Experiment
- It introduces an exogenous increase in police presence and displacement of population
- Measure the effect of the strike on distance of crime incidents to schools
- Overall number of crime incidents not significantly different.
- Examine the effect of strike on location of crime. Two possible ways in which strike could affect:
  - "Demand shock": Increased police presence at public schools may displace activity further away
  - "Supply shock": Less people/traffic in area may translate to lower opportunities for crime

Distributions: Public vs Private

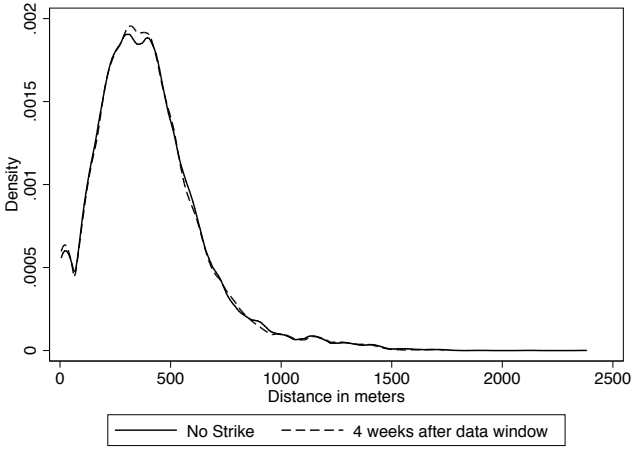
Public vs Private



Strike v No Strike (Public)



Data window vs 4 weeks after - Public



Econometric model

I estimate the following model:

$$Distance_i = \alpha + \beta Strike_i + \gamma t_i + \psi_d + \eta_c + \epsilon_i \tag{1}$$

- Framework has usual identification problems
- Will provide various specification tests and use distance to private schools as falsification tests.

Effect of strike on distances - by type of crime

Distance(in mts)	(1) Public	(2) Private	(3) Public	(4) Private	(5) Public	(6) Private
Strike(=1)	12.013	-6.161	20.348	2.722	9.802	-3.779
	[3.345]***	[4.705]	[5.929]***	[9.184]	[6.405]	[7.938]
Crime Type	All	All	Violent	Violent	Property	Property
Observations	26,765	26,765	7,342	7,342	12,269	12,269
R-squared	0.197	0.268	0.157	0.271	0.225	0.264

Specification test 1 - Alternative controls

Distance in meters:	(1)	(2)	(3)	(4)	(5)
Panel A: Public school					
Strike(=1)	12.013	11.999	8.928	9.291	12.741
	[3.345]***	[3.433]***	[3.042]***	[3.064]***	[4.680]***
Panel B: Private school					
Strike(=1)	-6.161	-6.085	-1.669	-1.846	-6.445
	[4.705]	[4.752]	[3.872]	[3.932]	[5.262]
Fixed Effects:	Com.	Com.*t	Beat	Beat*t	Com.
Other controls	-	-	-	-	Weather
Observations	26,765	26,765	26,765	26,765	26,765

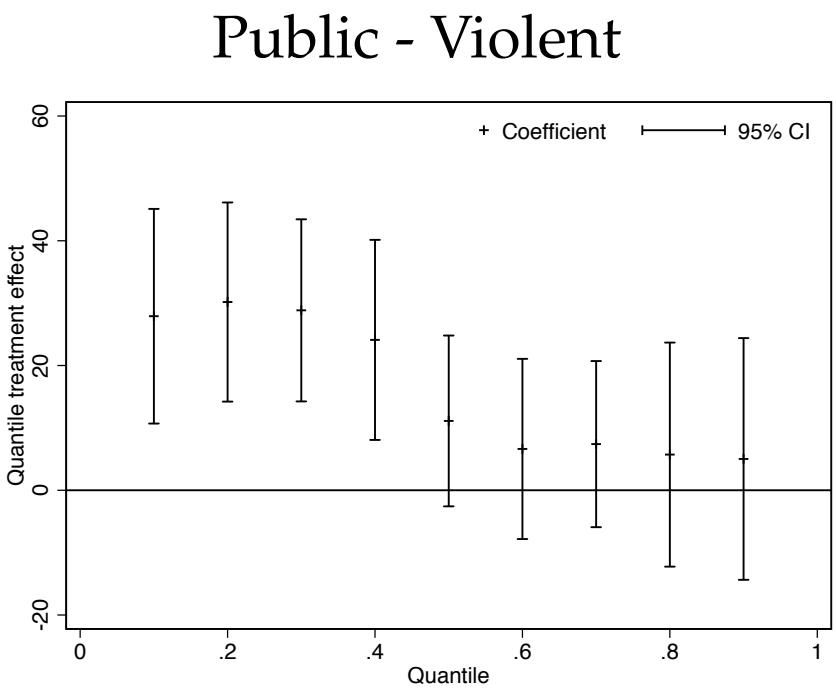
Specification test 2 - Data window selection

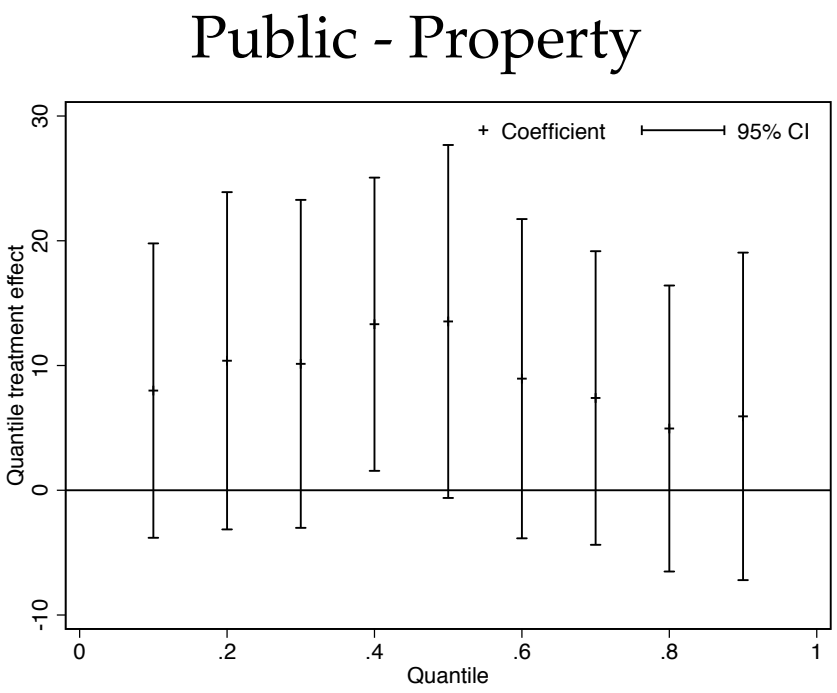
Distance in meters	(1)	(2)	(3)	(4)
Panel A: Public schools				
Strike(=1)	12.013	12.46	17.361	12.197
	[3.345]***	[3.563]***	[5.368]***	[3.363]***
Panel B: Private schools				
Strike(=1)	-6.161	-8.91	11.109	-7.071
	[4.705]	[4.917]*	[7.511]	[4.713]
Window	Original	- 2 weeks	No before	+ 2 weeks
Observations	26,765	19,723	23,106	33,647

Specification test 3 - Placebo test using data for 2011

Distance in meters:	(1) Public	(2) Private	(3) Public	(4) Private	(5) Public	(6) Private
Placebo Strike (=1)	-0.491	2.678	1.014	-6.977	-4.269	12.944
	[3.169]	[4.361]	[5.579]	[9.790]	[6.621]	[6.596]*
Crime type	All	All	Violent	Violent	Property	Property
Observations	28,939	28,939	8,218	8,218	13,041	13,041
R-squared	0.184	0.281	0.169	0.295	0.204	0.266

Quantile effects





Identifying the mechanism

- If police presence were only driving results, why is there no effects on property crimes?
- This could be explained by a displacement of people from school area.
- In the following section I will:
  - Explore different effects of distance to public high school vs elementary school
  - Explore heterogenous effects by school size

Differential effects on distance to HS vs Elementary

	(1)	(2)	(3)	(4)	(5)	(6)
Distance(mts):	Public high school			Public elementary school		
Strike(=1)	13.307	26.061	5.156	7.286	13.473	5.551
	[9.441]	[15.425]*	[12.793]	[3.408]**	[5.745]**	[6.680]
Crime type	All	Violent	Property	All	Violent	Property
Observations	26,765	7,342	12,269	26,765	7,342	12,269
R-squared	0.401	0.404	0.402	0.384	0.302	0.423

Heterogeneous effects by school size

Distance(mts):	(1) HS	(2) Elem.	(3) Private	(4) HS	(5) Elem.	(6) Private
Strike (=1)	-16.094	18.265	10.036	-2.019	7.614	5.482
	[26.671]	[9.858]*	[12.485]	[22.861]	[13.894]	[9.859]
Enrollment	0.07	0.062	0.065	0.071	0.094	0.052
	[0.049]	[0.033]*	[0.080]	[0.036]*	[0.055]*	[0.063]
Strike*Enroll	0.049	-0.009	-0.038	0.008	-0.003	-0.044
	[0.025]**	[0.014]	[0.033]	[0.023]	[0.023]	[0.040]
Crime type	Violent	Violent	Violent	Property	Property	Property
Observations	7,342	7,342	7,342	12,269	12,269	12,269
R-squared	0.408	0.304	0.272	0.406	0.427	0.265

Additional results

- No effects on time crime takes place (no temporal displacement)
- Coefficients do not change for truncating upper tail (expected after seeing quantiles)
- Next: compare these effects to those resulting from holidays/weekends to gain better insight on mechanisms.